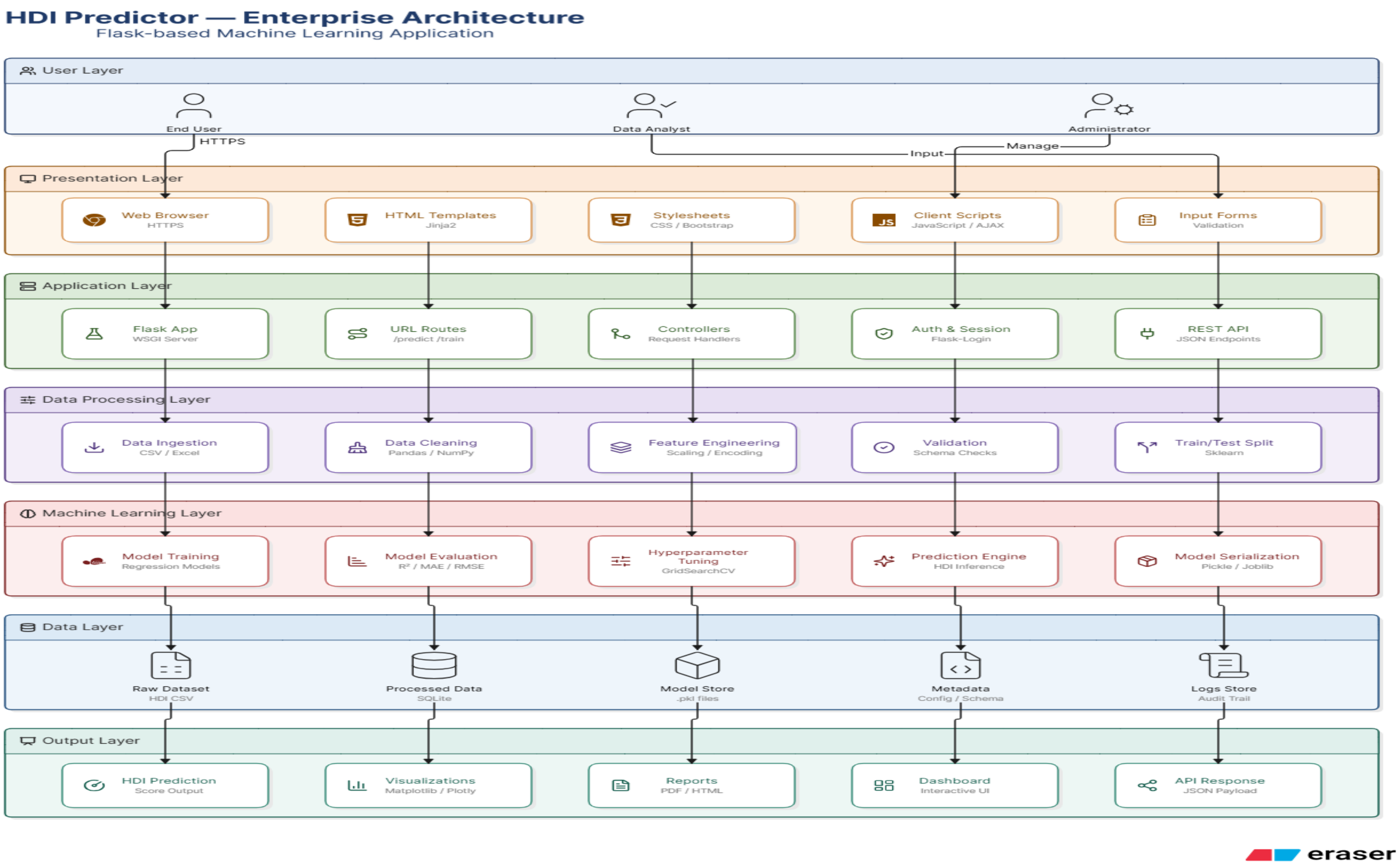
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	Human Development Index (HDI) Predictor
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode



REFERENCE:[REFERENCE IMAGE]

TABLE 1:

S.No	Component	Description	Technology
1.	User Interface	Interactive web interface	HTML, CSS, Bootstrap
2.	Application Logic	Request handling	Python, Flask
3.	Machine Learning	HDI prediction	Scikit-learn (Linear Regression)
4.	Data Processing	Cleaning & preprocessing	Pandas, NumPy
5.	Visualization	Graphs & charts	Matplotlib / Plotly
6.	Dataset	HDI dataset	CSV
7.	Storage	Local file storage	Local File System
8.	Deployment	Local development server	Flask Development Server

TABLE 2:

S.No	Characteristics	Description	Technology
1.	**Open Source Framework**	**Web framework**	**Flask**
2.	**Security**	**Input validation**	**Flask validation**
3.	**Scalable Architecture**	**Modular architecture**	**Flask + ML Modules**
4.	Availability	Local deployment	**Flask Server**
5.	Performance	Optimized prediction	Pandas + Scikit-learn

References:

[1] Flask Documentation. https://flask.palletsprojects.com/

[2] Scikit-learn Documentation. https://scikit-learn.org/stable/

[3] Pandas Documentation. https://pandas.pydata.org/docs/

[4] NumPy Documentation. https://numpy.org/doc/

[5] Matplotlib Documentation. https://matplotlib.org/stable/

[6] United Nations Development Programme (UNDP), Human Development Reports. https://hdr.undp.org/

[7] Human Development Data Center (UNDP).https://hdr.undp.org/data-center

[8] IBM Architecture Center. https://developer.ibm.com/components/architecture/

[9] Miro – Online Collaborative Whiteboard Platform https://miro.com/

[10] Eraser AI – Architecture Diagram Tool https://www.eraser.io/